(a) Prefix reuse

Saved interaction · Qwen3-4B

1. Build → 2. Append → 3. Edit early → **4. Restore**

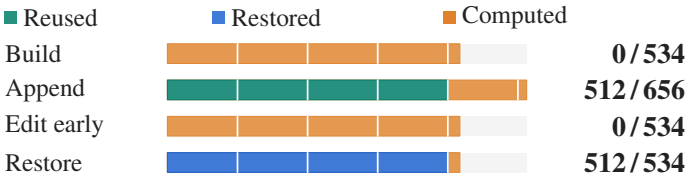
Prompt excerpt

```
You are maintaining a Python repository. Diagnose the timeout failure below.

deadline = monotonic() + timeout
connection = await client.acquire()
```

Prefix reuse

Reused / prompt tokens



Model response — Restore

The root cause is that the `timeout` parameter is only used to set a deadline for the function's execution, but it is not enforced during the `client.acquire()` call.

512 restored tokens · TTFT **47.3 ms** · DRAM → GPU

KV Inspector — After Build

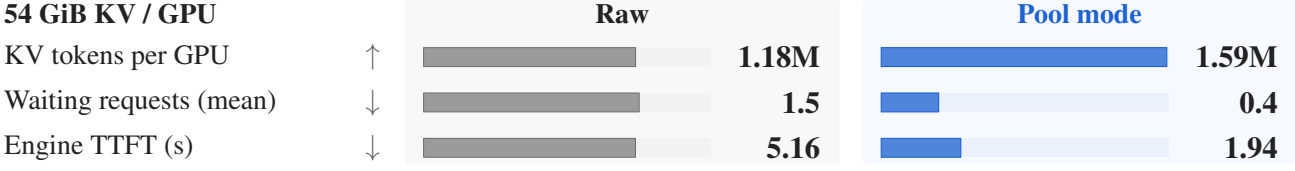
Independent export/import check
Block 1 · 128 tokens · 36 layers

BF16		Encoded blob
18.00 MiB	→	**12.73 MiB**

0 BF16 mismatches · 576 records

(b) GPU capacity

Recorded · $P = 64$ · $S = 12$ · NVMe 2 TiB



(c) History retention

Recorded · $P = 64$ · $S = 8$ · NVMe 640 GiB

